CRYPTOGRAPHY | FINAL PROJECT REPORT

CryptoChain Analyzer Dashboard

A live cryptographic and AI monitor for Bitcoin, Ethereum and Solana network health.

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1 Project Overview

CryptoChain Analyzer Dashboard is a Streamlit application that monitors three live networks: **Bitcoin**, **Ethereum** and **Solana**. The interface keeps the same M1–M8 structure for every asset, but adapts the meaning of each metric to the protocol being inspected. Bitcoin is used for classic Proof of Work and double-SHA-256; Ethereum focuses on EIP-1559 gas economics and proof-of-stake security assumptions; Solana focuses on slot finality, blockhash continuity and throughput stability.

Layer	Bitcoin	Ethereum	Solana
State	Difficulty, hash rate, block time	Base fee, gas used, block time	Slot, TPS, skip rate
Mechanism	80-byte header, nonce, target	Parent hash, gas target, next base fee	Blockhash, parent slot, finalized status
Evolution	Difficulty retargeting	Base-fee trend across recent blocks	TPS and slot-time trend
Security	Merkle proof, 51% attack cost	PoS stake-cost estimate	Skip-rate and finality score

2 Module Structure

The project follows the mandatory module sequence M1–M4 and extends it with advanced modules M5–M7. I also added **M8**, a custom risk-radar module, because the three chains expose different kinds of risk and a single visual score makes them easier to compare during a live demo. Some UI labels reuse the course template names; the table below gives the functional meaning used in the final multi-chain dashboard.

Mod.	Name	Role in the dashboard
M1	Live state monitor	Shows the current operating state: Bitcoin hash-rate and block time, Ethereum gas load, or Solana TPS and skip rate.
M2	Mechanism analyzer	Opens the protocol object behind the latest state: Bitcoin header fields, Ethereum fee-rule inputs, or Solana parent-slot data.
M3	Evolution history	Plots how the key metric changes over time: difficulty, base fee, or throughput.
M4	AI predictor	Forecasts the next relevant value for each asset using a transparent lightweight model.
M5	Verification module	Recomputes a concrete rule: Bitcoin Merkle inclusion, Ethereum next-base-fee update, or Solana parent-blockhash continuity.
M6	Security score	Translates protocol-specific risk into an interpretable score or curve: 51% attack risk, PoS capital-at-risk, or finality stability.
M7	Second AI method	Compares linear prediction with exponential smoothing and reports rolling MAE, so the model choice is evaluated.
M8	Custom risk radar	Extra module added by me: a five-factor radar (Security, Finality, Throughput, Fee Health, Stability), useful because BTC, ETH and SOL fail in different ways.

3 Cryptographic Metrics Displayed

The dashboard displays live evidence that each chain is preserving continuity, finality and economic security. The metrics are not identical because the protocols are different, but they answer the same cryptographic question: what makes the current state hard to rewrite or falsify?

3.1 Bitcoin: Proof of Work

For Bitcoin, M2 fetches the raw 80-byte block header and recomputes the double-SHA-256 block hash:

$$H = \mathrm{SHA256}(\mathrm{SHA256}(\mathrm{header})).$$

The result is reversed to Bitcoin display order. The parsed fields are `version`, previous block hash, `merkle_root`, timestamp, compact `bits` and nonce. The compact target is expanded as $T = c \cdot 2^{8(e-3)}$, and a block is accepted when $H \leq T$. M1 combines difficulty D with recent block time $\bar{t}$ to estimate hash rate:

$$R = \frac{D \cdot 2^{32}}{\bar{t}}.$$

M5 reconstructs a Merkle proof from transaction IDs. M6 implements Nakamoto's confirmation-risk model for an attacker with hash-rate fraction q after z confirmations [1, §11].

3.2 Ethereum: Gas and PoS Security

For Ethereum, the dashboard replaces mining difficulty with current block demand. M1 shows block height, base fee, gas used and block time. M2 displays the parent hash, gas limit, gas target and the protocol-computed next base fee. M5 recomputes the EIP-1559 update rule from `baseFeePerGas`, `gasUsed` and `gasLimit`; if the computed value matches the RPC value, the fee transition is marked verified [2]. M6 estimates economic security from assumed ETH price, total stake and slashing/loss percentage, exposing how proof-of-stake safety depends on capital at risk rather than on hash attempts.

3.3 Solana: Slot Continuity

For Solana, the relevant evidence is fast finality and slot-chain consistency. M1 shows finalized slot, TPS, skip rate and average block time. M2 displays the current blockhash, previous blockhash, parent slot, transaction count and confirmation depth. M5 verifies that the current slot's declared previous blockhash matches the parent slot's blockhash, which checks continuity of the finalized chain. M6 converts skip rate, confirmation depth and slot-time deviation into an operational security score. The Solana RPC commitment model defines `finalized` as the strongest confirmation state used by the network [3].

4 AI Component

The chosen AI approach is a **Predictor**. It forecasts the next relevant network variable for each asset: Bitcoin difficulty, Ethereum base fee and Solana TPS. M4 provides the main forecast view, while M7 compares a linear model with exponential smoothing. The common pipeline is:

Live API samples → Normalize time series → Linear + smoothing → Rolling MAE

Linear regression was kept because it is transparent: its slope explains direction and its intercept makes the prediction reproducible. Exponential smoothing is included because live crypto metrics often move in short bursts; it gives a second model that reacts more strongly to recent values. Both are lightweight enough to retrain on every refresh.

Asset	Predicted metric	Linear MAE	Smoothing MAE	Winner
BTC	Difficulty	3.62×10^{12}	1.16×10^{12}	Smoothing
ETH	Base fee (gwei)	0.0192	0.0120	Smoothing
SOL	TPS	100.09	77.31	Smoothing

The table was computed on 9 May 2026 with the live dashboard code. It is a dated evaluation snapshot, so exact MAE values can change when the live APIs are queried later. Each result uses a rolling one-step-ahead backtest: train on values before time t, predict t, record $|\hat{x}_t - x_t|$, and average the errors. In this run, exponential smoothing wins for all three assets, so the project reports model performance instead of assuming that the simpler linear trend is always best.

5 Conclusion

The final dashboard is therefore multi-chain, but still cryptographic in purpose. Bitcoin demonstrates hash-based Proof of Work and Merkle inclusion; Ethereum demonstrates rule-based fee verification and proof-of-stake economic security; Solana demonstrates slot finality and block-hash continuity. The AI component adds an evaluated forecasting layer on top of those metrics, making the dashboard both explanatory and operational.

References

[1] S. Nakamoto, *Bitcoin: A Peer-to-Peer Electronic Cash System*, 2008.
https://bitcoin.org/bitcoin.pdf

[2] V. Buterin et al., *EIP-1559: Fee market change for ETH 1.0 chain*, 2019.
https://eips.ethereum.org/EIPS/eip-1559

[3] Solana Foundation, *Solana RPC Overview*.
https://solana.com/docs/rpc